Roll Number:_____

Thapar Institute of Engineering & Technology, Patiala
Department of Electrical and Instrumentation Engineering
Auxiliary Examination

B.E.	Course Code: UBM605 Course Name: Data Structures and Algorithms
28th August 2023	Tuesday, 2.00 – 5.00 Hrs
Time: 3 Hours, M. Marks: 100	Name Of Faculty: Dr. Ashish Kumar Gupta, Dr. (Ms.) Mukesh Dalal

Note: All five questions are compulsory.

Q.1 a) a) What is a data structure and what is the need for a data structure? Differentiate between linear and non-linear data structure. (8)

b) What is an asymptotic analysis of an algorithm? Explain with the help of suitable graph and write the equation for all asymptotic notations (12)

Q.2 a) Write algorithms/pseudocode for the basic operations of stack using linked list. (12)

b) Write an algorithm for merge sort and sort the given values using Merge Sort: 67, 72, 77, 80, 85, 60, 58, 50, 45. Also, solve the running time equation of merge sort. (8)

Q.3 a) Consider the following max heap- 45, 30, 25, 15, 11, 8, 16. Insert a new node with value 57. Also, write the pseudocode for the insertion into the max heap. (8)

b) What is hashing? Explain the types of collision resolution techniques in hashing with suitable examples. (12)

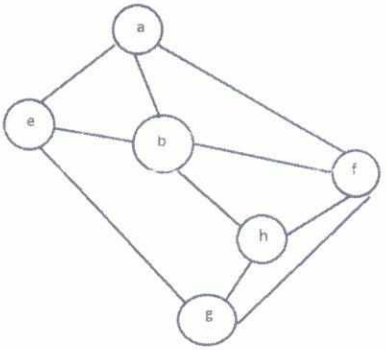
Q.4 a) What is a binary search tree? Suppose the numbers 6 , 5 , 1 , 8 , 3 , 4 , 0 , 9, 6, 2 are inserted in that order into an initially empty binary search tree. The binary search tree uses the usual ordering on natural numbers. Draw the binary search tree and what is the pre-order, in-order and postorder traversal sequence of the resultant tree? (10)

b) The preorder traversal sequence of a binary search tree is- (10)

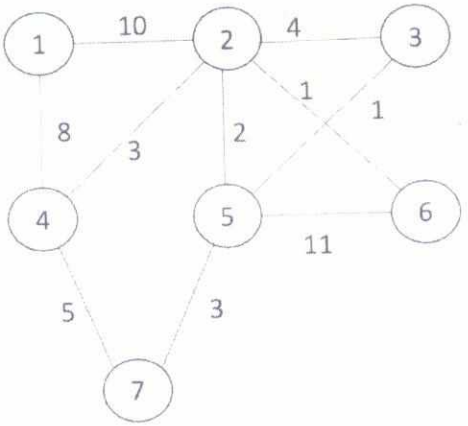
30 , 20 , 10 , 15 , 25 , 23 , 39 , 35 , 42

What is the postorder traversal sequence of the same tree?

Q.5 (a) Explain how graphs are represented in the memory? Consider the following graph and write the breadth first traversal of the following graph. Also, write the pseudocode for breadth first traversal in a graph. (10)



(b) What is the minimum spanning tree? Solve the below graph using kruskal's algorithm. (10)



Roll Number:_____

Thapar Institute of Engineering & Technology, Patiala
Department of Electrical and Instrumentation Engineering
Auxiliary Examination

B.E.	Course Code: UBM605 Course Name: Database Management Systems
31st August 2023	Thursday, 5.30 – 7:30 Hrs.
Time: 3 Hours, M. Marks: 100	Name Of Faculty: Dr. (Ms.) Mukesh Dalal

Note: All five questions are compulsory.

Q.1 a) What is a database management system? What are the problems of Hierarchical Model? Explain with examples. (10)

b) Discuss the importance of data independence used in DBMS as opposed to the File System. Also explain different types of data independence with suitable examples. (10)

Q.2 a) Suppose User 1 wants to fetch only Roll_Number and User 2 wants to fetch two Roll_Number and Name of all the students based on Table 1. Draw the three level architecture of RDBMS based on the given table. Also discuss any two points that show the comparison among different Data models. (10)

Table 1:Student_Details

Roll_Number	Name	Mobile Number	Email
1221	Akash	8985656561	akash@gmail.com
1222	Rita	8566006702	rita@gmail.com
1223	Akash	9416501114	akash01@gmail.com
1224	Sakshi	9585611121	sakshi@gmail.com

b) Discuss the differences between the following terms with appropriate examples: (10)
i) Left Outer Join and Right Outer Join.
ii) Trivial and Non Trivial Dependency.
iii) Generalization and Specialization.

Q.3 a) Consider the following functional dependencies in a database: (12)

Data_of_Birth → Age
Age → Eligibility
Name → Roll_number
Roll_number → Name
Course_number → Course_name
Course_number → Instructor
(Roll_number, Course_number) → Grade

The relation (Roll_number, Name, Data_of_Birth) is in which normal form? Explain your answer in detail by considering all the rejected and accepted FDs.

b) Discuss the difference/s and similarities between the third normal form and Boyce Coded Normal form. Also give suitable examples. (8)

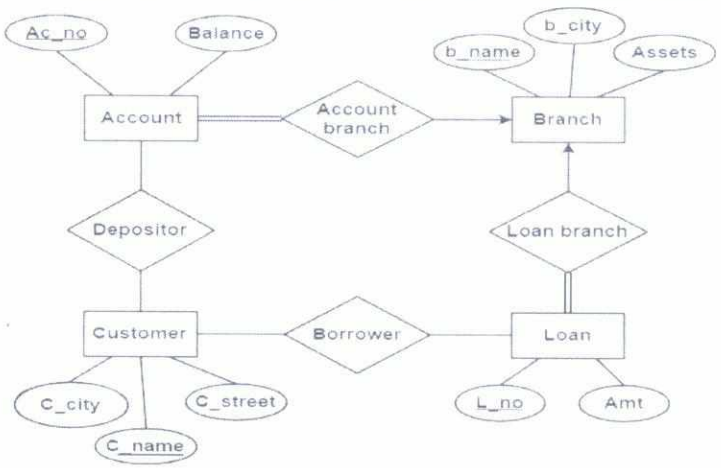
Q.4 a) What do you mean by Concurrency? Why is concurrency control required, discuss with suitable examples? Also, Consider the following schedules involving two transactions. (12)

S 1 : ri (X) ; r, (Y) ; r2(X) ; r2(Y) ; w2(Y) ; wi (X)
S2 : ri (X) ; r2(X) ; r2(Y) ; w2(Y) ; ri (Y) ; wi (X)

Which one of the above schedules are conflict serializable and why?

b) Discuss the different types of anomalies with suitable examples. (8)

Q.5 (a) Convert the following ER diagram into database tables. Explicitly mention the primary and foreign keys for each of the database tables. (10)



(b) Differentiate the following with suitable examples: (10)

i) DROP and DELETE
ii) ON DELETE SET NULL and ON DELETE CASCADE
iii) Entity integrity constraint and Referential Integrity constraint